



Letter

Family of quantum mutual information in multiparty quantum systems

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ABSTRACT

The characterization of information within a multiparty system is both significant and complex. This paper presents the concept of generalized conditional mutual information, along with a family of multiparty quantum mutual information measures. We provide interpretations and delineate the properties of these concepts, while also pointing out certain unresolved issues. The generalized conditional mutual information serves to encapsulate the interdependencies and correlations among various components of a multiparty quantum system. Additionally, various formulations of multiparty quantum mutual information contribute to a deeper comprehension of classical, quantum, and total correlations. These insights have the potential to propel fundamental research in the field of quantum information theory.

1. Introduction

In our information-driven universe, information is the cornerstone of communication, computation, and knowledge. Shannon in 1948 introduced the concept of *mutual information* which provided a framework for quantifying the amount of information shared between two systems, revolutionizing classical information theory. Classical mutual information $I(X : Y) = H(X) + H(Y) - H(X, Y)$, where $H(X)$ is the Shannon entropy, measures the amount of information obtained about one random variable through another random variable. It quantifies the reduction in uncertainty about one variable given knowledge of the other. In classical information theory [1], mutual information has diverse applications in data compression, error correction, and channel capacity, making it a fundamental tool in communication systems and coding theory.

This concept has now transcended into the quantum domain, playing a crucial role in quantum information theory [2–4] in particular. Quantum mutual information (QMI) is a generalization of classical mutual information to quantum systems. The QMI of a bipartite quantum state ρ_{AB} given by $I(A : B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$, where $S(\rho)$ is the von Neumann entropy, quantifies the total correlation [5,6], both classical and quantum, between subsystems A and B . It serves as a measure of correlation beyond entanglement. QMI is essential in multiple areas of quantum information processing like quantum communication, quantum computing, and quantum cryptography [2–4]. It is especially important in quantifying quantum channel capacities [7,8]. In quantum machine learning [9,10], it quantifies the information exchanged be-

tween various representations of quantum datasets. It is also significant as a probe for many-body localization [11], and in quantifying quantum objectivity [12].

Understanding various correlations [13–16] in systems involving more than two parties becomes increasingly important. Quantum entanglement and correlations are resources which may be shared among multiple parties but with certain restrictions. Thus, in measuring correlations of multipartite systems, some inequalities on distribution of correlations arise naturally—say, the monogamy inequalities [17–19]. Although the problem of entanglement generation and detection in multipartite systems has been well studied [20–22], there still exist interesting problems. For example, the formalism to distinguish quantum correlation from classical correlation is not straightforward. In the present study, we consider quantum mutual information in multiparty systems, provide interpretations of their quantifiers, and present some inequalities.

In classical information theory, mutual information between multiple random variables can be straightforwardly extended. In the quantum realm, however, correlations are more complex due to the exotic quantum phenomena such as nonlocality, superposition and measurement problem. Multiparty quantum mutual information (MQMI) extends the concept of bipartite QMI to systems with multiple parties, offering insights into the intricate correlations that arise in quantum states. It seeks to quantify the total correlations among several subsystems within a quantum state, and has implications for understanding and leveraging the correlations in multiparty quantum systems. Mutual information

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and related measures [5,6,23–29] are famed measures of multipartite information and correlation. The journey of mutual information from classical to quantum domains underscores its profound elegance and significance. As quantum technologies continue to advance, the significance of mutual information in both the realms becomes increasingly evident. The two main contributions of this paper are:

1. Concept of *generalized conditional mutual information* which is an extension of the conditional entropy $S(A|B) = S(\rho_{AB}) - S(\rho_B)$ and conditional mutual information $I(A : B|C) = S(\rho_{AC}) + S(\rho_{BC}) - S(\rho_C) - S(\rho_{ABC})$. This quantity can encapsulate every possible interdependency and correlation of any subsystem of a multipartite system.
2. Introduction of a family of multipartite quantum mutual information. There are at least $n - 1$ MQMI for an n (≥ 2)-party quantum system. Moreover, any positive linear combination of these MQMIs is another MQMI. Some linear combinations of these MQMIs can yield negative values.

It is evident that multiple expressions of MQMI arise due to different ways of defining and quantifying correlations in complex quantum systems. Each expression may capture unique aspects of these correlations, leading to diverse applications and insights. A straightforward consequence of multiple expressions of MQMI would be enhanced understanding of classical, quantum and total correlations. By providing a multifaceted understanding of correlations in multipartite quantum systems, multiple expressions of MQMI can drive fundamental research in quantum information theory. By providing different perspectives on correlations and their operational interpretations, we can explore new theoretical models and deepen our understanding of quantum systems.

This paper is organized as follows. In Sec. 2, we consider the preliminaries such as notation and definitions. The notion of *generalized conditional mutual information* which is the generalization of conditional entropy and conditional mutual information to multipartite systems is introduced in Sec. 3. In Sec. 4, we introduce a family of multipartite quantum mutual information and provide their interpretations and properties. We also speculate them to be *secrecy monotones* which are useful in cryptography. Finally, we conclude and discuss some unresolved issues in Sec. 5.

2. Setup

2.1. Preliminaries

We consider a multipartite quantum system $X_1 X_2 \cdots X_n$ represented by finite dimensional density matrix $\rho_{X_1 X_2 \cdots X_n} \in \mathcal{H}_1^{d_1} \otimes \mathcal{H}_2^{d_2} \otimes \cdots \otimes \mathcal{H}_n^{d_n}$. The reduced density matrix of subsystem $\bar{X}$ is obtained by partial tracing over the remaining subsystems $\bar{X}$: $\rho_{\bar{X}} = \text{tr}_{\bar{X}}(\rho_{X\bar{X}})$. The von Neumann entropy (in bits) given by

$$S(\rho) := -\text{tr}(\rho \log_2 \rho) = -\sum_i \lambda_i \log_2 \lambda_i, \quad (1)$$

where $\lambda_i \geq 0$ and $\sum_i \lambda_i = 1$, is the quantum counterpart of Shannon entropy. Shannon entropy is the average information content of a probability distribution. The von Neumann entropy satisfies the inequalities [30,31]: $S(\rho_X) - S(\rho_Y) \leq S(\rho_{XY}) \leq S(\rho_X) + S(\rho_Y) \leq S(\rho_{XZ}) + S(\rho_{YZ})$ and $S(\rho_Y) + S(\rho_{XYZ}) \leq S(\rho_{XY}) + S(\rho_{YZ})$. We denote the von Neumann entropy of subsystem $X_i X_j$ as $S(\rho_{X_i X_j}) \equiv S(X_i X_j) \equiv S_{ij}$, and so on. $S(\rho) = 0$ for a pure quantum state. Another important entropy for our purpose is quantum relative entropy (QRE) which measures the *closeness* of two density matrices. It is defined as

$$D(\tau||\sigma) := \text{tr}(\tau \log_2 \tau) - \text{tr}(\tau \log_2 \sigma), \quad (2)$$

if $\text{supp}(\tau) \subseteq \text{supp}(\sigma)$, and infinity otherwise. The *support* of a Hermitian matrix is the Hilbert space spanned by its eigenvectors with nonzero

eigenvalues [2]. The QRE is monotonic under partial trace, completely positive and trace-preserving (CPTP) maps, and positive maps [31,32]. Let $\mathcal{X} = \{X_1, X_2, \dots, X_n\}$, $[n] = \{1, 2, \dots, n\}$, and $\mathbf{1} = 1$ such that $S(\mathbf{1}) = 0$ and for any system X , $\rho_X \mathbf{1} = \rho_X = \mathbf{1} \rho_X$. [One can alternatively consider $\mathbf{1} = \text{diag}\{1, 1, \dots, 1\}$ with the requirement that its dimension is self-adjusting!]

2.2. Correlations and monotones

Let us consider a function $f(\rho_{X_1 \cdots X_n})$ defined on $\rho_{X_1 \cdots X_n}$ and mention below some plausible and useful properties.

- (P1) Symmetry: $f(\rho_{X_1 \cdots X_n})$ is symmetric under the interchange of any two parties X_j and X_k .
- (P2) Semipositivity: $f(\rho_{X_1 \cdots X_n}) \geq 0$.
- (P3) Vanishing on product states: $f(\rho_{X_1} \otimes \cdots \otimes \rho_{X_n}) = 0$.
- (P4) Monotonicity under some local operations [local (completely) positive maps].
- (P5) Monotonicity under classical communications (public announcement or communication over phone).
- (P6) Additivity: $f(\rho \otimes \sigma) = f(\rho) + f(\sigma)$ and $f(\rho^{\otimes n}) = n f(\rho)$.
- (P7) Continuity: $f(\rho)$ is a continuous (smooth) function of its argument ρ .

If f satisfies (P2) semipositivity and (P3) vanishing on product density matrices, it is a measure of the amount of *correlation* between the parties. A nonnegative correlation function that observes (P4) monotonicity under local operations and (P5) monotonicity under classical communications is called a *monotone*. A *secrecy monotone*, in addition to (P2–P5), satisfies (P6) additivity and (P7) continuity. If the secrecy monotone is expected to measure the amount of information (secrecy) shared by the communicating parties $\{X_1, X_2, \dots, X_n\}$ with the hostile party Eve, then the following properties are natural [33,34]:

- (P8) Monotonicity under local operations by Eve.
- (P9) Monotonicity under classical (public) communication by Eve.

Whether the information in question is correlation, monotone or secrecy monotone should be clear from its properties and the context.

3. Generalized conditional mutual information

In this section, we introduce the notion of generalized conditional mutual information (GCMI). It is the multipartite extension of conditional entropy $S(A|B) = S(AB) - S(B)$ and conditional mutual information $I(A : B|C) = S(AC) + S(BC) - S(C) - S(ABC)$. It can encompass every possible interdependency or correlation (interaction information) of any subsystem of a multipartite system. We define the GCMI as the information contained in subsystems $X_{k_1} X_{k_2} \cdots X_{k_m}$ of a multipartite system $X_1 X_2 \cdots X_n$ ($n \geq m$) but not in Y , where Y (acting as a single system) is either $\mathbf{1}$ or one or more remaining subsystems,

$$I(X_{k_1} : X_{k_2} : \cdots : X_{k_m} | Y) := -S(Y) + \sum_{j=1}^m (-1)^{j+1} \sum_{k_1 < \cdots < k_j \in [m]} S(X_{k_1} X_{k_2} \cdots X_{k_j} Y). \quad (3)$$

A few remarkable points about $I(X_{k_1} : X_{k_2} : \cdots : X_{k_m} | Y)$ are as follows.

1. $I(X_{k_1} : X_{k_2} | Y) \geq 0$. This follows from the strong subadditivity of von Neumann entropy.
2. $I(X_{k_1} X_{k_2} \cdots X_{k_m} | \mathbf{1}) = S(X_{k_1} X_{k_2} \cdots X_{k_m})$.
3. $I(X_1 : X_2 : \cdots : X_n | \mathbf{1})$ is the information (correlation) common to subsystems $X_1, X_2, \dots$, and X_n .
4. It can assume negative values [35–39].

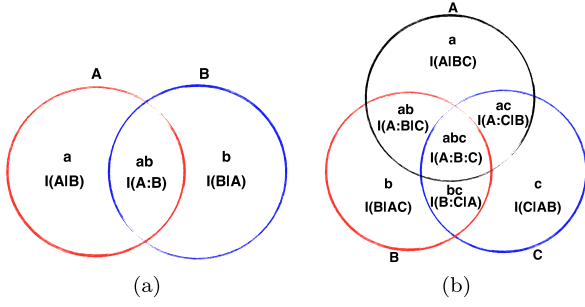


Fig. 1. (a) Two-variable and (b) three-variable Venn diagrams with possible intersecting regions and generalized conditional mutual information. Here $S(X) \equiv I(X|\mathbf{1})$ is the information content of subsystem X , the information contained in subsystem X but not in subsystem Y is given by the conditional entropy $S(X|Y) = S(XY) - S(Y) = S(X) - I(X:Y) \equiv I(X|Y)$, and $I(X:Y)$ is the information between X and Y .

In the above, $I(X|\mathbf{1}) \equiv I(X)$, $I(X_1 : X_2 : X_3|\mathbf{1}) \equiv I(X_1 : X_2 : X_3)$, and so on. In multipart systems, however, we prefer to keep the notation $I(X|\mathbf{1})$ rather than $I(X)$ to remind us the fact that either $I(X) \equiv I(X|\mathbf{1}) = -S(\mathbf{1}) + S(X)$ is analogous to conditional information or $I(X)$ is not a multipart (total) correlation.

The nontriviality in characterization of information and correlations begins to emerge with three-party system onwards. We illustrate the idea of GCMi using the tripartite system ABC represented by density matrix ρ_{ABC} [see Fig. 1(b)].

1. Information in A is $[a + ab + ac + abc] = S(\rho_A) \equiv I(A|\mathbf{1})$.
2. Information in A but neither in B nor in C (i.e., information strictly contained in A) is $[a] = -S(\rho_{BC}) + S(\rho_{ABC}) \equiv I(A|BC)$.
3. Information in A (and possibly in B) but not in C is $[a + ab] = -S(\rho_C) + S(\rho_{AC}) \equiv I(A|C)$.
4. Information common to A and B (and possibly in C) is $[ab + abc] = S(\rho_A) + S(\rho_B) - S(\rho_{AB}) \equiv I(A : B|\mathbf{1})$.
5. Information common to A and B but not in C is $[ab] = -S(\rho_C) + S(\rho_{AC}) + S(\rho_{BC}) - S(\rho_{ABC}) \equiv I(A : B|C)$.
6. Information common to A , B and C is $[abc] = S(\rho_A) + S(\rho_B) + S(\rho_C) - S(\rho_{AB}) - S(\rho_{AC}) - S(\rho_{BC}) + S(\rho_{ABC}) \equiv I(A : B : C|\mathbf{1})$.
7. Information in ABC (as a single system) is $S(\rho_{ABC}) \equiv I(ABC|\mathbf{1})$.

Indeed, there are several possibilities. Different arrangements or configurations of subsystems yield, in general, different values of information or correlation.

8. Three-party total correlations amongst A , B and C are given by

$$\begin{aligned}
 T_3(A : B : C) &= S(\rho_A) + S(\rho_B) + S(\rho_C) - S(\rho_{ABC}) \\
 &= D(\rho_{ABC} \| \rho_A \otimes \rho_B \otimes \rho_C) \\
 &= I(A : BC) + I(B : C) \\
 &= I(A : B|C) + I(A : C|B) + I(B : C|A) + 2I(A : B : C|\mathbf{1}), \\
 &\text{and} \\
 S_3(A : B : C) &= S(\rho_{AB}) + S(\rho_{AC}) + S(\rho_{BC}) - 2S(\rho_{ABC}) \\
 &= I(A : BC) + I(B : C|A) \\
 &= I(A : B|C) + I(A : C|B) + I(B : C|A) + I(A : B : C|\mathbf{1}) \\
 &= S(\rho_{ABC}) - I(A|BC) - I(B|AC) - I(C|AB).
 \end{aligned}$$

4. Family of MQMI

The bipartite QMI $I(A : B) = S(A) + S(B) - S(AB)$ is the measure of total correlation (classical and quantum) between subsys-

tems A and B [5,6], satisfies the Araki-Lieb inequality $I(A : B) \leq 2 \min\{S(\rho_A), S(\rho_B)\}$ [30], is invariant under local unitary operations, and is nonincreasing under tracing out a subsystem. Here we introduce a family of entropic functions constructed on $\rho_{X_1 \dots X_n}$, and show below that they serve as n -party quantum mutual information.

4.1. Definition

We define the k^{th} ($1 \leq k \leq n$) multipart quantum mutual information on $\rho_{X_1 \dots X_n}$ as

$$\begin{aligned}
 M_k^{(n)}(X_1 : X_2 : \dots : X_n) \\
 := \sum_{j_1 < \dots < j_k \in [n]} S(X_{j_1} \dots X_{j_k}) - \binom{n-1}{k-1} S(X_1 \dots X_n).
 \end{aligned} \quad (4)$$

The superscript “ n ” in $M_k^{(n)}$ denotes the number of subsystems (single or composite) separated by colons, and the coefficients $\binom{n-1}{k-1}$ for given n constitute the n^{th} -row of the Pascal’s triangle. We posit a few remarks here. First, the choice of definition in Eq. (4) is motivated by the fact that the two well-established MQMI [28,33] in the literature are members of this family. Second, this multipart quantum mutual information contains all two and more parties interactions, as described in [28], and not just the n -party interaction $I(X_1 : X_2 : \dots : X_n|\mathbf{1})$. Third, $M_1^{(2)}(X : Y) = I(X : Y) = I(X : Y|\mathbf{1})$.

For $p + q = n$, one can consider p -party versus q -party partitions of $\mathcal{X}$ such that

$$M_p^{(n)} + M_q^{(n)} = \sum_{k\bar{k} \in \{p|q \text{ partitions of } \mathcal{X}\}} I(\rho_k : \rho_{\bar{k}}). \quad (5)$$

See Appendix A for more relations between $M_k^{(n)}$.

The existence of several expressions for multipart quantum mutual information can have profound consequences and diverse applications in quantum information theory and technology. As research in this area continues to evolve, the insights gained from multiple MQMI measures will be instrumental in unlocking the full potential of quantum information science. The following observations, for instance, merit attention.

1. $M_k^{(n)}$ are independent multipart quantum mutual information. As these yield distinct values for an arbitrary state, the parties must make *a priori* choice about which multipart correlation they want to consider. This *a priori* choice should be obvious (self-revealing) once their interpretations are known.
2. To maximize their correlation, parties should consider the quantity $M_k^{(n)}$ where $k = n/2$ (for even n) or $k = (n \pm 1)/2$ (for odd n). For minimal correlation, $M_1^{(n)}$ and $M_{n-1}^{(n)}$ are good candidates.
3. Corresponding to this family of multipart quantum mutual information, one can introduce a family of multipart quantum discord. Discord is the difference between *unmeasured* total correlation and *measured* total correlation present in a quantum state.
4. Information deviation due to a map or channel Φ (this may include noise) acting on a multipart quantum state ρ is $|\mathcal{Q}(\Phi(\rho)) - \mathcal{Q}(\rho)|$, where $\mathcal{Q}$ is either $M_k^{(n)}$, $M^{(n)}$, $C^{(n)}$, or generalized conditional mutual information.
5. Suppose the eavesdropper E interacts with the system ρ_{AB} . Then, the desirable condition for the secure secret sharing of information between A and B is that the sum of correlations of E with A and B should be minimal (refer to Fig. 1(b) with E in place of C). That is,

$$\begin{aligned}
 &I(A : E|B) + I(B : E|A) + I(A : B : E) \\
 &= M_2^{(3)}(A : B : E) - I(A : B|E) \\
 &= I(AB : E) \rightarrow 0.
 \end{aligned} \quad (6)$$

A more stringent condition would be that each of $I(A : E|B)$, $I(B : E|A)$, and $I(A : B : E)$ is either zero or tends to zero.

6. $M_k^{(n)}$ together with $\sum_k c_k M_k^{(n)}$ can be used to distinguish quantum states.

The various equivalent expressions of two eminent MQMIs [28,33] are:

$$T_n(X_1 : X_2 : \dots : X_n) \equiv M_1^{(n)}(X_1 : X_2 : \dots : X_n) \\ = \sum_{k=1}^n S(\rho_{X_k}) - S(\rho_{X_1 \dots X_n}) \quad (7)$$

$$= D(\rho_{X_1 \dots X_n} \| \rho_{X_1} \otimes \dots \otimes \rho_{X_n}) \quad (8)$$

$$= \sum_{k=1}^{n-1} I(X_k : X_{k+1} \dots X_n) \quad (9)$$

$$= \sum_{j=2}^n (j-1) \sum_{k_1 < \dots < k_j \in [n]} I(X_{k_1} : \dots : X_{k_j} | X_{k_{j+1}} \dots X_{k_n}), \quad (10)$$

and

$$S_n(X_1 : X_2 : \dots : X_n) \equiv M_{n-1}^{(n)}(X_1 : X_2 : \dots : X_n) \\ = \sum_{k=1}^n S(\rho_{X_1 \dots X_{k-1} X_{k+1} \dots X_n}) - (n-1) S(\rho_{X_1 \dots X_n}) \quad (11)$$

$$= I(X_1 : X_2 \dots X_n) + \sum_{k=2}^{n-1} I(X_k : X_{k+1} \dots X_n | X_1 \dots X_{k-1}) \quad (12)$$

$$= \sum_{j=2}^n \sum_{k_1 < \dots < k_j \in [n]} I(X_{k_1} : \dots : X_{k_j} | X_{k_{j+1}} \dots X_{k_n}) \quad (13)$$

$$= S(\rho_{X_1 \dots X_n}) - \sum_{k=1}^n I(X_k | \overline{X_k}). \quad (14)$$

Both T_n and S_n measure both the classical and the quantum correlations. T_n and S_n are referred to as “total correlation” [5,23] and “dual total correlation” [24,25] respectively, quantum secrecy monotones [33], and multipart quantum mutual information [28] from the information-theoretic point of view.

We further find that $M_k^{(n)}$, for fixed k ($1 \leq k < n$), is nondecreasing under discarding of any one party or grouping together any two parties (see Appendix B). That is,

$$M_k^{(n)}(X_1 : \dots : X_n) \geq M_k^{(n-1)}(X_1 : \dots : X_{n-1}), \quad (15)$$

$$M_k^{(n)}(X_1 : \dots : X_n) \geq M_k^{(n-1)}(X_1 : \dots : X_{n-1} X_n). \quad (16)$$

4.2. Interpretations

Interpretations of $T_n = M_1^{(n)}$.— From Eq. (8), T_n is interpreted as the minimal relative entropy between $\rho_{X_1 \dots X_n}$ and a product density matrix $\sigma_{X_1} \otimes \dots \otimes \sigma_{X_n}$ (the minimum being attained when $\sigma_{X_k} = \rho_{X_k}$). From Eq. (9), T_n is the sum of decorrelation costs when the parties decorrelate themselves locally one by one from the rest using $I(X_k : X_{k+1} \dots X_n)$ bits of randomness [5]. From Eq. (10), T_n is the sum of $(j-1)$ -times all $(j \geq 2)$ -party interactions.

Interpretations of $S_n = M_{n-1}^{(n)}$.— S_n like T_n , from Eq. (12), can be interpreted as the sum of decorrelation costs when the parties decorrelate themselves locally one by one from the rest using $I(X_k : X_{k+1} \dots X_n | X_1 \dots X_{k-1})$ bits of randomness. S_n , using Eq. (13), is the sum of all interactions of two and more parties only once [28]. Equivalently, from Eq. (14), it is the entropy of the whole system less the sum of information in nonintersecting regions.

Nevertheless, while all $M_k^{(n)}$ can be viewed as the sum of two and more parties interactions, not all of them have straightforward physical

Table 1

Values of multipart quantum mutual information $M_k^{(n)}$ and common information $C^{(n)}$ of generalized Greenberger-Horne-Zeilinger states $|gGH Z_n\rangle = \frac{1}{\sqrt{p}} |0\rangle^{\otimes n} + e^{i\phi} \sqrt{1-p} |1\rangle^{\otimes n}$, Dicke states $|D_n^r\rangle = \frac{1}{\sqrt{\binom{n}{r}}} \sum_p P[|0\rangle^{\otimes n-r} |1\rangle^{\otimes r}]$,

three-qutrit totally antisymmetric state $|\psi_{as}\rangle = \frac{1}{\sqrt{6}}(|123\rangle - |132\rangle + |231\rangle - |213\rangle + |312\rangle - |321\rangle)$, four-qubit cluster state $|C_4\rangle = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle - |1111\rangle)$, and $|HS_4\rangle = \frac{1}{\sqrt{6}}(|0011\rangle + |1100\rangle + \omega(|1010\rangle + |0101\rangle) + \omega^2(|1001\rangle + |0110\rangle))$. Here $h(p) := -p \log_2 p - (1-p) \log_2 (1-p)$ is the binary entropy, $\omega = e^{\frac{2\pi i}{3}}$, and “ $\times$ ” stands for *not applicable*. Note that $C^{(n)}$ can be negative and $C^{(n=odd)}$ vanishes for pure states.

State	$M_1^{(n)}$	$M_2^{(n)}$	$M_3^{(n)}$	$M_4^{(n)}$	$C^{(n)}$
$ gGH Z_2\rangle$	$2 h(p)$	0	$\times$	$\times$	$2 h(p)$
$ gGH Z_3\rangle$	$3 h(p)$	$3 h(p)$	0	$\times$	0
$ D_3^1\rangle$	2.75489	2.75489	0	$\times$	0
$ \psi_{as}\rangle$	4.75489	4.75489	0	$\times$	0
$ gGH Z_4\rangle$	$4 h(p)$	$6 h(p)$	$4 h(p)$	0	$2 h(p)$
$ D_4^1\rangle$	3.24511	6	3.24511	0	0.490225
$ D_4^2\rangle$	4	7.50978	4	0	0.490225
$ C_4\rangle$	4	10	4	0	-2
$ HS_4\rangle$	4	10.75489	4	0	-2.75489
$ gGH Z_5\rangle$	$5 h(p)$	$10 h(p)$	$10 h(p)$	$5 h(p)$	0
$ D_5^1\rangle$	3.60964	9.70951	9.70951	3.60964	0
$ D_5^2\rangle$	4.85475	12.95462	12.95462	4.85475	0

or operational meaning like $M_1^{(n)}$ and $M_{n-1}^{(n)}$. Therefore, any operational interpretation of these mutual information would be appreciated.

4.3. Properties

Here we discuss a number of useful properties that $M_k^{(n)}$ satisfy.

- $M_k^{(n)}$ is invariant under local unitary operations because von Neumann entropy is invariant under unitary operations.
- $M_1^{(n)} = T_n$ and $M_{n-1}^{(n)} = S_n$ satisfy the following properties [28, 33]: (P1–P9) above, (P10) $S_n = T_n$ for pure states, and (P11) $S_n(\rho_{X_1 X_2 \dots X_n}) \leq T_n(\rho_{X_1 X_2 \dots X_n}) + 2S(\rho_{X_1 X_2 \dots X_n})$.
- $M_n^{(n)} = 0$ (as expected) because $(X_1 X_2 \dots X_n)$ as a single system has no mutual information.
- Suppose $1 \leq p, q \leq n-1$ such that $p \neq q$ and $p+q=n$. Then $M_p^{(n)} = M_q^{(n)}$ for pure states and hence can be called “dual” to each other. We envisage from Table 1 that for pure states the profile of $M_k^{(n)}$ versus k is analogous to a truncated Gaussian (a point or a straight line being the particular case).
- $M_k^{(n)}$ satisfy (P1) symmetry, (P3) vanishing on product states, (P6) additivity and (P7) continuity. These properties are also satisfied by any quantity which is a linear combination of $M_k^{(n)}$. In particular,

$$M^{(n)}(X_1 : X_2 : \dots : X_n) \\ = \sum_{k=1}^n \lambda_k M_k^{(n)}(X_1 : X_2 : \dots : X_n), \quad (17)$$

where $\lambda_k \geq 0$ and $\sum_k \lambda_k = 1$. We conjecture below that $M_k^{(n)}$ and $M^{(n)}$ are plausible candidates for secrecy monotones. Another important linear combination of $M_k^{(n)}$ is

$$C^{(n)}(X_1 : X_2 : \dots : X_n) \\ = \sum_{k=1}^n (-1)^{k+1} M_k^{(n)}(X_1 : X_2 : \dots : X_n), \quad (18) \\ = I(X_1 : X_2 : \dots : X_n | \mathbf{1}).$$

$C^{(n)}$ is the information (correlation) common to all X_k s. It can, however, be negative (see Table 1). Moreover, it vanishes identically for pure odd-party quantum states:

$$C^{(n=\text{odd})} = \sum_{k=\text{odd}} M_k^{(n)} - \sum_{k=\text{even}} M_k^{(n)} = 0. \quad (19)$$

6. For pure states, $C^{(n)} \leq M_k^{(n)} = M_{n-k}^{(n)}$. For mixed states, we expect them to obey the inequality:

$$C^{(n)} \leq \left(M_1^{(n)} \approx M_{n-1}^{(n)} \right) < \left(M_2^{(n)} \approx M_{n-2}^{(n)} \right) < \dots < M_{n/2}^{(n)}. \quad (20)$$

We also surmise that for $k_1 < k_2 \leq n/2$ and $c \geq c_{k_2, k_1} = \frac{k_2 \binom{n}{k_2}}{k_1 \binom{n}{k_1}}$,

$$c M_{k_1}^{(n)} \geq M_{k_2}^{(n)}. \quad (21)$$

7. $M_k^{(n)}(X_1 : X_2 : \dots : X_n)$ in Eq. (4) is semipositive.

Proof. It is obvious for pure states because $S(X_1 X_2 \dots X_n)$ vanishes identically. For mixed states, the semipositivity of $M_1^{(n)}$, $M_{n-1}^{(n)}$, and $M_{k=n/2}^{(n)}$ (for even n) follows from Eqs. (8), (9), Eq. (12), and Eq. (5) respectively. In general, $M_k^{(n)}$ is semipositive because it is nondecreasing under discarding a subsystem: $M_k^{(n)} \geq M_k^{(n-1)} \geq \dots \geq M_k^{(k)} = 0$ [see Eq. (15)]. $M_k^{(n)}$ can also be shown nonnegative using the subadditivity and the strong subadditivity of von Neumann entropy (see Appendix C). The idea is to eliminate $S(X_1 X_2 \dots X_n)$ terms. This can be achieved by grouping together (repeatedly) two appropriate entropy terms of small number of parties to obtain an entropy term having greater number of parties. One should, however, take care that while grouping no entropy term, except $S(X_1 X_2 \dots X_n)$, appears twice or more. ■

Thus, $M_k^{(n)}$ in Eq. (4) and $M^{(n)}$ in Eq. (17) satisfy a number of useful properties: symmetry, semipositivity, vanishing on product states, additivity, and continuity. Hence, they constitute a family of multiparty quantum mutual information.

4.4. Conjecture and remark

Secrecy monotones quantify the amount of *secret correlation* shared by the parties of a multipartite system. They are useful in the study of quantum (as well as classical) cryptography. $M_1^{(n)} = T_n$ and $M_{n-1}^{(n)} = S_n$ satisfy properties (P4, P5) and (P8, P9), and qualify for secrecy monotones [33]. We speculate (see Appendix D for argument) that $M_k^{(n)}$ ($k = 2, 3, \dots, n-2$) in Eq. (4) and $M^{(n)}$ in Eq. (17) also meet the criteria (P4, P5) and (P8, P9) for being considered as measures of secrecy monotone.

In addition to these n -party symmetric monotones, we also have other monotones $M_k^{(m)}$ ($1 \leq k \leq m < n$) on n -party quantum states $\rho_{X_1 \dots X_n}$ which can be obtained by grouping together any two or more of the n -parties. All these monotones, however, are not all linearly independent.

5. Conclusions

In this study, we have conducted a comprehensive analysis of entropy-based information within multiparty systems. Firstly, we introduced the concept of generalized conditional mutual information. Next, we presented a family of multiparty quantum mutual information, which is anticipated to significantly contribute to fundamental research in quantum information theory. This advancement is expected to enhance our comprehension of classical, quantum, and total correlations, and consequences thereof. Notably, this framework includes the two well-established multiparty quantum mutual information measures.

While various interpretations exist for $M_1^{(n)}$ and $M_{n-1}^{(n)}$, the interpretation of other measures $M_k^{(n)}$ remains unclear. Therefore, some operational interpretations of these mutual information measures would be beneficial. Additionally, we conjecture that the remaining measures $M_k^{(n)}$ and $M^{(n)} = \sum_k \lambda_k M_k^{(n)}$, where $\lambda_k \geq 0$ and $\sum_k \lambda_k = 1$, will exhibit monotonicity under local operations and classical communication, thereby qualifying as secrecy monotones. We also posit that our formalism will be instrumental in characterizing measures of multiparty nonclassical correlations.

CRedit authorship contribution statement

Asutosh Kumar: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Relations between $M_k^{(n)}$

1. The recurrence relation(s) for T_n and S_n are as follows:

$$T_n(X_1 : X_2 : \dots : X_n) = T_{n-1}(X_{k_1} : X_{k_2} : \dots : X_{k_{n-1}}) + I(X_{k_1} \dots X_{k_{n-1}} : X_{k_n}), \quad (A.1)$$

$$T_n(X_1 : X_2 : \dots : X_n) = T_{n-1}(X_{k_1} X_{k_2} : X_{k_3} : \dots : X_{k_n}) + I(X_{k_1} : X_{k_2}), \quad (A.2)$$

and

$$S_n(X_1 : X_2 : \dots : X_n) = S_{n-1}(X_{k_1} X_{k_2} : X_{k_3} : \dots : X_{k_n}) + I(X_{k_1} : X_{k_2} | X_{k_3} \dots X_{k_n}), \quad (A.3)$$

where X_{k_j} belongs to and exhaust the set $\mathcal{X}$. It is evident that the recurrence relation is not unique. For the choice of $\{X_{k_{n-j}} = X_{j+1}\}_{j=0}^{n-1}$, Eq. (A.1) and Eq. (A.2) separately yields Eq. (9), and Eq. (A.3) yields Eq. (12).

2. Relations between $M_k^{(n_1)}$ and $M_{k+1}^{(n_2)}$.

$$M_2^{(n)} + \sum_{j < k \in [n]} I(X_j : X_k) = (n-1) M_1^{(n)}, \quad (A.4)$$

$$(k+1) M_{k+1}^{(n)} + \sum_{j_1 < \dots < j_k \in [n]} \sum_{i \neq j_1 \neq \dots \neq j_k} I(X_{j_1} \dots X_{j_k} : X_i) = (n-k) M_k^{(n)} + \left(1 - \frac{k}{n}\right) \binom{n}{k} M_1^{(n)}, \quad (A.5)$$

$$M_{n-1}^{(n)}(X_1 : \dots : X_n) = M_{n-2}^{(n-1)}(X_1 : \dots : X_{n-1}) + \sum_{k=1}^{n-1} \left(S_{\overline{X_k}} + S_{\overline{X_n}} - S_{X_1 \dots X_n} - S_{\overline{X_k X_n}} \right). \quad (A.6)$$

Appendix B. Proof of Eqs. (15), (16)

Here we show that $M_k^{(n)}$, for fixed k ($1 \leq k < n$), is nondecreasing under discarding of any one party or grouping together any two parties. That is,

$$M_k^{(n)}(X_1 : X_2 : \dots : X_n) \geq M_k^{(n-1)}(X_1 : X_2 : \dots : X_{n-1}), \quad (B.1)$$

$$M_k^{(n)}(X_1 : X_2 : \dots : X_n) \geq M_k^{(n-1)}(X_1 : X_2 : \dots : X_{n-1} X_n). \quad (B.2)$$

Above inequalities follow from the definition of $M_k^{(n)}$ and the entropic inequalities $S(X) + S(Y) \geq S(XY)$ (subadditivity) and $S(XY) + S(YZ) \geq S(Y) + S(XYZ)$ (strong subadditivity), as shown below. First, we show that $M_k^{(n)}$ is nondecreasing under discarding a subsystem.

$$\begin{aligned}
& M_1^{(n)}(X_1 : X_2 : \dots : X_n) \\
&= \sum_{k=1}^n S_k - S_{12\dots n} \\
&= \sum_{k=1}^{n-1} S_k - S_{12\dots(n-1)} + (S_{12\dots(n-1)} + S_n - S_{12\dots n}) \\
&= M_1^{(n-1)}(X_1 : X_2 : \dots : X_{n-1}) + I(X_1 X_2 \dots X_{n-1} : X_n | 1) \\
&= M_1^{(n-1)}(X_1 : X_2 : \dots : X_{n-1}) + M_1^{(2)}(X_1 X_2 \dots X_{n-1} : X_n), \\
&\geq M_1^{(n-1)}(X_1 : X_2 : \dots : X_{n-1}), \\
& M_2^{(n)}(X_1 : X_2 : \dots : X_n) \\
&= \sum_{j < k \in [n]} S_{jk} - (n-1)S_{12\dots n} \\
&= \sum_{j < k \in [n-1]} S_{jk} - (n-2)S_{12\dots(n-1)} \\
&\quad + \left(\sum_{j=1}^{n-1} S_{jn} + (n-2)S_{12\dots(n-1)} - (n-1)S_{12\dots n} \right) \\
&\geq M_2^{(n-1)}(X_1 : X_2 : \dots : X_{n-1}) + \left(\sum_{j=1}^{n-2} S_j + S_{(n-1)n} - S_{12\dots n} \right) \\
&\geq M_2^{(n-1)}(X_1 : X_2 : \dots : X_{n-1}) \\
&\quad + M_1^{(n-1)}(X_1 : X_2 : \dots : X_{n-2} : X_{n-1} X_n), \\
&\geq M_2^{(n-1)}(X_1 : X_2 : \dots : X_{n-1}), \\
& M_3^{(n)}(X_1 : X_2 : \dots : X_n) \\
&= \sum_{i < j < k \in [n]} S_{ijk} - \binom{n-1}{2} S_{12\dots n} \\
&= \sum_{i < j < k \in [n-1]} S_{ijk} - \binom{n-2}{2} S_{12\dots(n-1)} \\
&\quad + \left(\sum_{i < j \in [n-1]} S_{ijn} + \binom{n-2}{2} S_{12\dots(n-1)} - \binom{n-1}{2} S_{12\dots n} \right) \\
&\geq M_3^{(n-1)}(X_1 : X_2 : \dots : X_{n-1}) \\
&\quad + \left(\sum_{j=2}^{n-1} S_{1jn} + \sum_{i=2}^{n-2} \sum_{j=i+1}^{n-1} S_{ij} - (n-2)S_{12\dots n} \right) \\
&\geq M_3^{(n-1)}(X_1 : X_2 : \dots : X_{n-1}).
\end{aligned}$$

Similarly, one can show that $M_{k>3}^{(n)}(X_1 : X_2 : \dots : X_n) \geq M_{k>3}^{(n-1)}(X_1 : X_2 : \dots : X_{n-1})$. Next, we show that $M_k^{(n)}$ is nondecreasing under grouping together two parties, specifically for $n=4$.

$$\begin{aligned}
M_1^{(4)}(X_1 : X_2 : X_3 : X_4) &= (S_1 + S_2 + S_3 + S_4) - S_{1234} \\
&= (S_1 + S_2 + S_{34} - S_{1234}) + (S_3 + S_4 - S_{34}) \\
&= M_1^{(3)}(X_1 : X_2 : X_3 X_4) + M_1^{(2)}(X_3 : X_4), \\
&\geq M_1^{(3)}(X_1 : X_2 : X_3 X_4), \\
M_2^{(4)}(X_1 : X_2 : X_3 : X_4) \\
&= S_{12} + S_{13} + S_{14} + S_{23} + S_{24} + S_{34} - 3S_{1234} \\
&= (S_{12} + S_{134} + S_{234} - 2S_{1234}) \\
&\quad + (S_{13} + S_{14} + S_{23} + S_{24} + S_{34} - S_{134} - S_{234} - S_{1234})
\end{aligned}$$

$$\begin{aligned}
&\geq M_2^{(3)}(X_1 : X_2 : X_3 X_4) + (S_1 + S_2 + S_{34} - S_{1234}), \\
&= M_2^{(3)}(X_1 : X_2 : X_3 X_4) + M_1^{(3)}(X_1 : X_2 : X_3 X_4), \\
&\geq M_2^{(3)}(X_1 : X_2 : X_3 X_4), \\
M_3^{(4)}(X_1 : X_2 : X_3 : X_4) &= S_{123} + S_{124} + S_{134} + S_{234} - 3S_{1234} \\
&\geq 0 = M_3^{(3)}(X_1 : X_2 : X_3 X_4). \quad \blacksquare
\end{aligned}$$

Appendix C. Semipositivity of $M_k^{(n)}$

Here we illustrate the nonnegativity of $M_k^{(n)}$ for $n=5$.

$$\begin{aligned}
M_1^{(5)} &= (S_1 + S_2 + S_3 + S_4 + S_5) - S_{12345} \geq S_{12345} - S_{12345} = 0. \\
M_2^{(5)} &= S_{12} + S_{13} + S_{14} + S_{15} + S_{23} + S_{24} + S_{25} + S_{34} \\
&\quad + S_{35} + S_{45} - 4S_{12345} \\
&= (S_{12} + S_{34}) + S_{15} + (S_{13} + S_{45}) + S_{24} + (S_{14} + S_{23}) \\
&\quad + S_{25} + S_{35} - 4S_{12345} \\
&\geq (S_{1234} + S_{15}) + (S_{1345} + S_{24}) + (S_{1234} + S_{25}) + S_{35} - 4S_{12345} \\
&\geq (S_{12345} + S_1) + (S_{12345} + S_4) + (S_{12345} + S_2) + S_{35} - 4S_{12345} \\
&= (S_1 + S_2 + S_4) + S_{35} - S_{12345} \\
&\geq (S_{124} + S_{35}) - S_{12345} \\
&\geq S_{12345} - S_{12345} = 0. \\
M_3^{(5)} &= S_{123} + S_{124} + S_{125} + S_{134} + S_{135} + S_{145} + S_{234} + S_{235} \\
&\quad + S_{245} + S_{345} - 6S_{12345} \\
&= (S_{123} + S_{145}) + (S_{124} + S_{235}) + (S_{135} + S_{234}) + (S_{134} + S_{245}) \\
&\quad + (S_{125} + S_{345}) - 6S_{12345} \\
&\geq (S_1 + S_{12345}) + (S_2 + S_{12345}) + (S_3 + S_{12345}) + (S_4 + S_{12345}) \\
&\quad + (S_5 + S_{12345}) - 6S_{12345} \\
&= (S_1 + S_2 + S_3 + S_4 + S_5) - S_{12345} \geq 0. \\
M_4^{(5)} &= (S_{1234} + S_{1235}) + (S_{1245} + S_{1345}) + S_{2345} - 4S_{12345} \\
&\geq (S_{123} + S_{12345}) + (S_{145} + S_{12345}) + S_{2345} - 4S_{12345} \\
&\geq (S_{123} + S_{145}) + S_{2345} - 2S_{12345} \\
&\geq (S_1 + S_{12345}) + S_{2345} - 2S_{12345} \\
&= (S_1 + S_{2345}) - S_{12345} \\
&\geq S_{12345} - S_{12345} = 0.
\end{aligned}$$

Alternatively, one can also endeavor to obtain a recurrence relation for $M_k^{(n)}$ which expresses it as a positive sum of bipartite mutual information $I(A : B)$ and conditional mutual information $I(A : B|C)$. Then, the semipositivity of $M_k^{(n)}$ is trivial. We, however, note that obtaining the recurrence relation is neither unique [see Eqs. (A.1), (A.2), (A.3)] nor easy. For example,

$$\begin{aligned}
& M_2^{(n)}(X_1 : X_2 : \dots : X_n) \\
&= M_2^{(n-1)}(X_1 : X_2 : \dots : X_{n-1}) + M_1^{(n-1)}(X_1 : X_2 : \dots : X_{n-1}) \\
&\quad + I(X_n : X_2 \dots X_{n-1} | X_1) \\
&\quad + \sum_{j=2}^{n-1} I(X_n : X_1 \dots X_{j-1} X_{j+1} \dots X_{n-1} | X_j). \quad \blacksquare \tag{C.1}
\end{aligned}$$

Appendix D. Argument for secrecy monotones of $M_k^{(n)}$

Our argument for the speculation that $M_k^{(n)}$ meet the criteria of secrecy monotones is as follows. Consider a five-party quantum system $\mathcal{X} = \{X_1, \dots, X_5\}$, for example. We know that the von Neumann entropy is invariant under unitary transformations including the permutation or particle exchange operator. That is, $S_{12345} = S_{13245} = S_{23145}$, etc. Then

$$\begin{aligned} & M_2^{(5)}(X_1 : X_2 : \dots : X_5) \\ &= S_{12} + S_{13} + S_{14} + S_{15} + S_{23} + S_{24} + S_{25} + S_{34} + S_{35} \\ & \quad + S_{45} - 4S_{12345} \\ &= (S_{12} + \log_2(d_3 d_4 d_5) - S_{12345}) \\ & \quad + (S_{13} + \log_2(d_2 d_4 d_5) - S_{12345}) + \dots \\ & \quad + (S_{45} + \log_2(d_1 d_2 d_3) - S_{12345}) + 6S_{12345} - \log_2(d_1 d_2 d_3 d_4 d_5)^6 \\ &= D(\rho_{12345} || \rho_{12} \otimes I_{345}) + D(\rho_{12345} || \rho_{13} \otimes I_{245}) + \dots \\ & \quad + D(\rho_{12345} || I_{123} \otimes \rho_{45}) + 6S_{12345} - \log_2(d_1 d_2 d_3 d_4 d_5)^6, \end{aligned}$$

and

$$\begin{aligned} & M_3^{(5)}(X_1 : X_2 : \dots : X_5) \\ &= S_{123} + S_{124} + S_{125} + S_{134} + S_{135} + S_{145} + S_{234} + S_{235} \\ & \quad + S_{245} + S_{345} - 6S_{12345} \\ &= (S_{123} + \log_2(d_4 d_5) - S_{12345}) + (S_{124} + \log_2(d_3 d_5) - S_{12435}) + \dots \\ & \quad + (S_{345} + \log_2(d_1 d_2) - S_{12345}) + 4S_{12345} - \log_2(d_1 d_2 d_3 d_4 d_5)^4 \\ &= D(\rho_{12345} || \rho_{123} \otimes I_{45}) + D(\rho_{12345} || \rho_{124} \otimes I_{35}) + \dots \\ & \quad + D(\rho_{12345} || I_{12} \otimes \rho_{345}) + 4S_{12345} - \log_2(d_1 d_2 d_3 d_4 d_5)^4. \end{aligned}$$

In the last steps above, the multiplicative factor of total entropy $S_{12\dots n}$ or the exponent of $\prod_{i=1}^n d_i$ can be obtained as $\binom{n}{k} \left(1 - \frac{k}{n}\right)$. While $D(\rho_{AB} || \rho_A \otimes \rho_B)$ is always well-behaved, $D(\rho_{AB} || \rho_B \otimes \rho_A)$ is not in general due to the support condition. Therefore, in the second steps above, the parties (subsystems) have been rearranged beforehand to resolve the support condition of QRE. Note that the operations such as local quantum operations and local measurements and public classical communication can be regarded as local positive maps [33]. Because $D(\rho || \sigma) \geq D(\Phi(\rho) || \Phi(\sigma))$ for any local positive map Φ , it implies that $M_k^{(n)}$ are monotonic under local quantum operations and classical communications. ■

Data availability

No data was used for the research described in the article.

References

- [1] T. Cover, J. Thomas, *Elements of Information Theory*, John Wiley & Sons, 1991.
- [2] M.A. Nielsen, I.L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press, 2000.
- [3] M.M. Wilde, *Quantum Information Theory*, Cambridge University Press, 2013.
- [4] J. Preskill, *Lecture Notes for Physics 229: Quantum Information and Computation*, CreateSpace Independent Publishing Platform, 2015.
- [5] B. Groisman, S. Popescu, A. Winter, Quantum, classical, and total amount of correlations in a quantum state, *Phys. Rev. A* 72 (2005) 032317.
- [6] N. Li, S. Luo, Total versus quantum correlations in quantum states, *Phys. Rev. A* 76 (2007) 032327.
- [7] C.H. Bennett, P.W. Shor, Quantum channel capacities, *Science* 303 (2004) 1784–1787.
- [8] A.S. Holevo, Quantum channel capacities, *Quantum Electron.* 50 (2020) 440.
- [9] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, S. Lloyd, Quantum machine learning, *Nature* 549 (2017) 195–202.
- [10] G. Carleo, I. Cirac, K. Cranmer, L. Daudet, M. Schuld, N. Tishby, L. Vogt-Maranto, L. Zdeborová, Machine learning and the physical sciences, *Rev. Mod. Phys.* 91 (2019) 045002.
- [11] G.D. Tomasi, S. Bera, J.H. Bardarson, F. Pollmann, Quantum mutual information as a probe for many-body localization, *Phys. Rev. Lett.* 118 (2017) 016804.
- [12] D.A. Chisholm, L. Innocenti, G.M. Palma, Importance of using the averaged mutual information when quantifying quantum objectivity, *Phys. Rev. A* 110 (2024) 012218.
- [13] I. Bengtsson, K. Życzkowski, *Geometry of Quantum States: An Introduction to Quantum Entanglement*, Cambridge University Press, 2006.
- [14] R. Horodecki, P. Horodecki, M. Horodecki, K. Horodecki, Quantum entanglement, *Rev. Mod. Phys.* 81 (2009) 865.
- [15] K. Modi, A. Brodutch, H. Cable, T. Paterek, V. Vedral, The classical-quantum boundary for correlations: discord and related measures, *Rev. Mod. Phys.* 84 (2012) 1655.
- [16] A. Bera, T. Das, D. Sadhukhan, S.S. Roy, A. Sen(De), U. Sen, Quantum discord and its allies: a review of recent progress, *Rep. Prog. Phys.* 81 (2018) 024001.
- [17] S.-Y. Liu, Y.-R. Zhang, L.-M. Zhao, W.-L. Yang, H. Fan, General monogamy property of global quantum discord and the application, *Ann. Phys.* 348 (2014) 256–269.
- [18] A. Kumar, Conditions for monogamy of quantum correlations in multipartite systems, *Phys. Lett. A* 380 (38) (2016) 3044–3050.
- [19] H.S. Dhar, A.K. Pal, D. Rakshit, A. Sen(De), U. Sen, Monogamy of quantum correlations – a review, in: *Lectures on General Quantum Correlations and Their Applications*, in: Part of the Series Quantum Science and Technology, Springer International Publishing, 2017, pp. 23–64.
- [20] C. Song, et al., Generation of multicomponent atomic Schrödinger cat states of up to 20 qubits, *Science* 365 (6453) (2019) 574–577.
- [21] C.-T. Chen, et al., ScQ cloud quantum computation for generating Greenberger-Horne-Zeilinger states of up to 10 qubits, *Sci. China, Phys. Mech. Astron.* 65 (2022) 110362.
- [22] K. Xu, et al., Metrological characterization of non-Gaussian entangled states of superconducting qubits, *Phys. Rev. Lett.* 128 (2022) 150501.
- [23] S. Watanabe, Information theoretical analysis of multivariate correlation, *IBM J. Res. Dev.* 4 (1) (1960) 66–81.
- [24] T.S. Han, Linear dependence structure of the entropy space, *Inf. Control* 29 (1975) 337–368.
- [25] T.S. Han, Nonnegative entropy measures of multivariate symmetric correlations, *Inf. Control* 36 (1978) 133–156.
- [26] Z. Walczak, Total correlations and mutual information, *Phys. Lett. A* 373 (2009) 1818–1822.
- [27] G.L. Giorgi, B. Bellomo, F. Galve, R. Zambrini, Genuine quantum and classical correlations in multipartite systems, *Phys. Rev. Lett.* 107 (2011) 190501.
- [28] A. Kumar, Multiparty quantum mutual information: an alternative definition, *Phys. Rev. A* 96 (2017) 012332.
- [29] Sk Sazim, P. Agrawal, Quantum mutual information and quantumness vectors for multiqubit systems, *Quantum Inf. Process.* 19 (2020) 216.
- [30] A. Wehrl, General properties of entropy, *Rev. Mod. Phys.* 50 (1978) 221.
- [31] M.B. Ruskai, Inequalities for quantum entropy: a review with conditions for equality, *J. Math. Phys.* 43 (2002) 4358–4375.
- [32] A. Müller-Hermes, D. Reeb, Monotonicity of the quantum relative entropy under positive maps, *Ann. Henri Poincaré* 18 (2017) 1777–1788.
- [33] N.J. Cerf, S. Massar, S. Schneider, Multipartite classical and quantum secrecy monotones, *Phys. Rev. A* 66 (2002) 042309.
- [34] M. Horodecki, P. Horodecki, R. Horodecki, Limits for entanglement measures, *Phys. Rev. Lett.* 84 (2000) 2014.
- [35] N.J. Cerf, C. Adami, Negative entropy and information in quantum mechanics, *Phys. Rev. Lett.* 79 (1997) 5194–5197.
- [36] M. Horodecki, J. Oppenheim, A. Winter, Partial quantum information, *Nature* 436 (2005) 673–676.
- [37] M. Horodecki, J. Oppenheim, A. Winter, Quantum state merging and negative information, *Commun. Math. Phys.* 269 (2007) 107–136.
- [38] L. del Rio, J. Aberg, R. Renner, O. Dahlsten, V. Vedral, The thermodynamic meaning of negative entropy, *Nature* 474 (2011) 61–63.
- [39] G. Gour, M.M. Wilde, S. Brandtsen, I.J. Geng, Inevitability of knowing less than nothing, arXiv:2208.14424 [quant-ph].